

Exp 6-Finite Word length Effects

6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

PROGRAM

```
clc;
clear;
close all;
%% Signal
fs = 1000; f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);
%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};
figure;
for i = 1:length(L)
    % Quantization
    xq{i} = round((x+1)*(L(i)-1)/2)*(2/(L(i)-1)) - 1;
    % Error & MSE
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);
    % Plot Quantized Signals
    subplot(4,1,i+1);
```

```

stairs(t,xq{i},colors{i},'LineWidth',1.2);
title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
grid on;
end
% Original Signal
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal'); grid on;
% Display MSE
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));
%% Error Plots
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t,e{i});
    title(['Error (' num2str(L(i)) ' Levels)']);
    grid on;
end

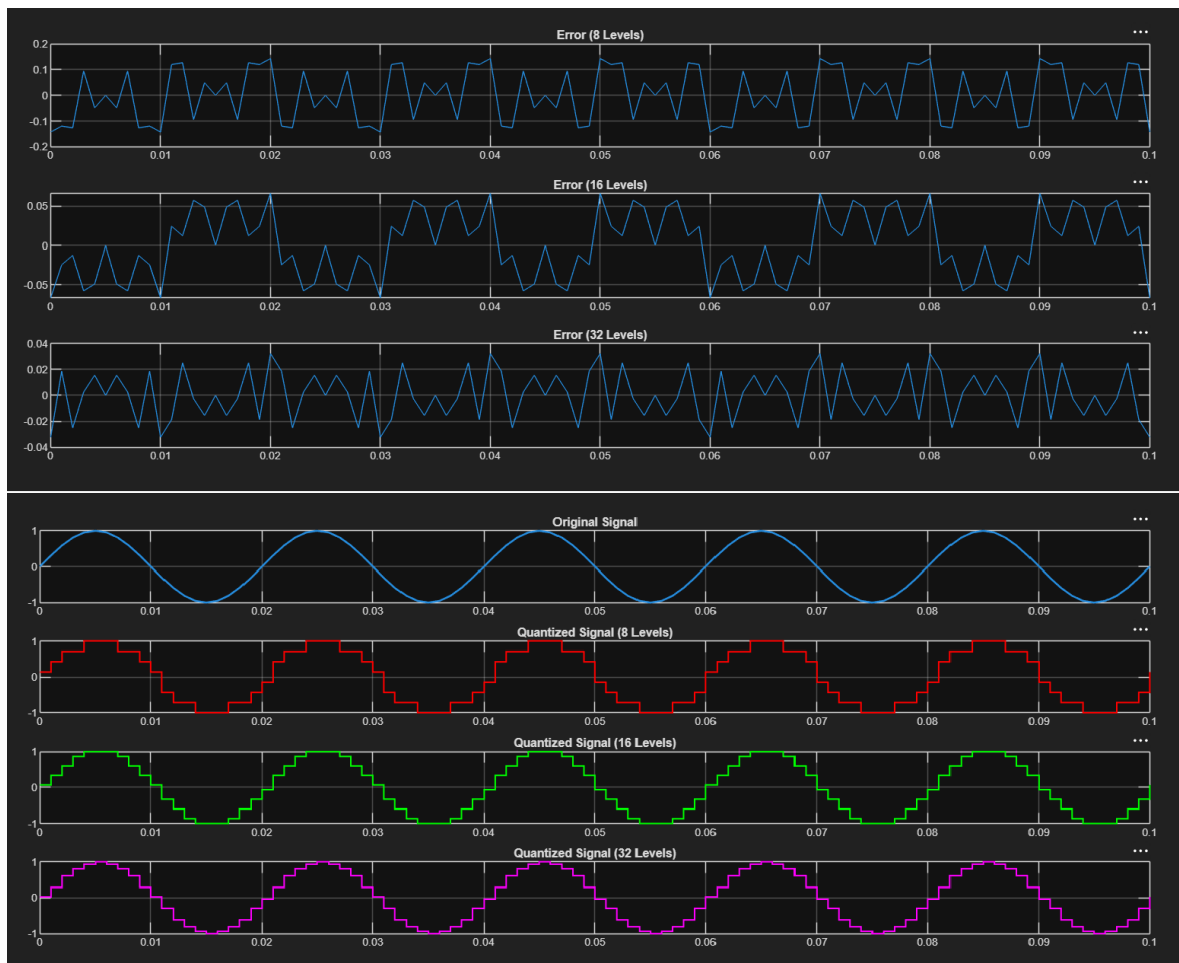
```

OUTPUT

```

MSE for 8 Levels = 0.010473
MSE for 16 Levels = 0.001763
MSE for 32 Levels = 0.000357

```



6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation using Rounding Techniques in MATLAB

PROGRAM

```

clc; clear; close all
n=0:79;
x=sin(2*pi*n/40);
q=8;
xt=floor(x*q)/q;
xr=round(x*q)/q;
et=x-xt;
er=x-xr;

```

% Fig 1

figure

plot(n,x,'b',n,xr,'r','LineWidth',1.5)

grid on

title('Original vs Quantized')

legend('Original','Quantized')

% Fig 2

figure

plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)

grid on

title('Truncation vs Rounding')

legend('Original','Truncated','Rounded')

% Fig 3

figure

stem(n,et,'r')

hold on

stem(n,er,'g')

grid on

title('Quantization Error')

legend('Truncation','Rounding')

% Fig 4

figure

bar([mean(et.^2) mean(er.^2)])

set(gca,'XTickLabel',{'Truncation','Rounding'})

grid on

title('MSE Comparison')

ylabel('MSE')

OUTPUT

